

Joshua Cowell

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EDUCATION

Bachelor of Science in Mechanical Engineering

Liberty University, Lynchburg, VA

Expected Graduation: May 2027

GPA - 3.82

PROFESSIONAL EXPERIENCE

Thermal Hydraulics Engineering Intern, Framatome Nuclear

May 2025 - Dec 2025

- Developed a predictive wear mathematical model in Python for critical reactor components, proving to be over 75% more accurate compared to the previous model.
- Programmed a Python algorithm using a 6-dimensional Xarray to calculate safe reactor operating conditions, reducing computation time from >800 hours to minutes compared to prior methods.
- Performed system-level flow and vibration analyses using AFT Fathom and custom FIV modeling software.
- Awarded \$2,500 for my final summer presentation and hired on part-time during the school year.

Chief Systems Engineer, Liberty Rocketry (9th/156 international teams in 2024)

June 2025 - Present

- Led a 60-member student team in the design, testing, and launch readiness of a high-powered rocket.
- Oversaw all engineering decisions across propulsion, avionics, recovery, and aerodynamic subsystems.
- Facilitated communication between internal teams, external vendors, and university faculty to optimize project efficiency.
- Displayed advanced proficiency in engineering software (e.g., SolidWorks, ANSYS, MATLAB) to verify the accuracy and feasibility of all subsystem models and simulations.

Lead Undergraduate Research Assistant, Liberty University (CERE)

Sep 2024 - Present

- Led a team of 5 undergraduate students to perform split-Hopkinson pressure bar (SHPB) test, including sample preparation, high-speed imaging, electron microscopy.
- Assisted in the design and manufacturing of custom SHPB systems to fit material testing needs.

Senior Structural Engineer, Liberty Rocketry

Sep 2023 - June 2025

- Utilized ANSYS Fluent & Mechanical, RocketPy, OpenRocket, and Onshape for simulations, FEA, and computational modeling, ensuring compliance with competition standards.
- Delivered reports to guarantee the rocket's compliance with competition standards and flight requirements.

Camp Counselor Woodlands Camp, GA

May 2024 - Aug 2024

- Provided 24-hour care during weeklong outdoor camping trips with 12 children per week (ages 8 to 18), teaching survival skills, rock climbing, whitewater rafting, horseback riding, and cave camping.

TECHNICAL SKILLS

- Software: AFT Fathom, ANSYS fluent, ANSYS mechanical, SolidWorks, Onshape, OpenRocket, RocketPy
- Machines: Scanning Electron Microscope (SEM), Metal X 3D Printer, CNC, Waterjet, Lathe
- Programming Languages: MATLAB, Python

PROJECTS

Unmanned Aerial Vehicle (UAV)

- Engineered a custom UAV with 3D-printed parts to support customizability and quick repairs.
- Conducted ANSYS CFD simulation to validate control surface performance.

Rocketry

- Built a two-stage rocket that reached over 2,500 feet and utilized my custom recovery system.